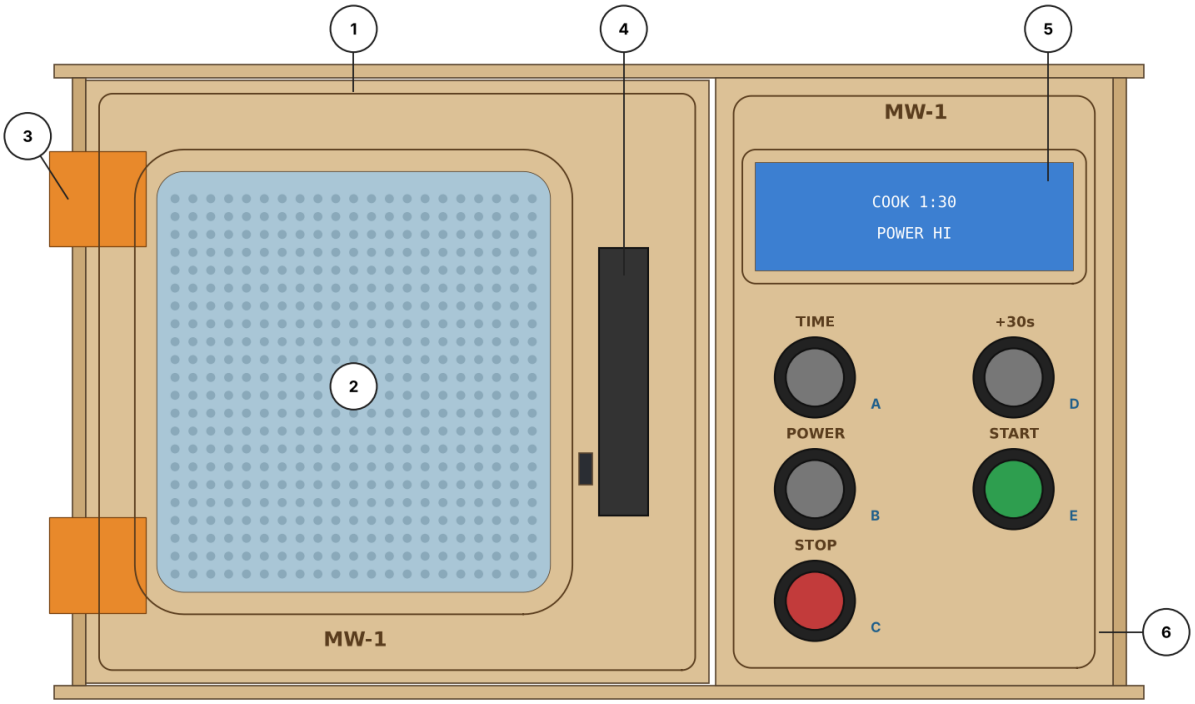
MW-1 Mock Microwave

User Manual

A countertop microwave stand-in for robot testing. It does not heat.



Read this manual before first use. Keep it with the unit.

Rev 1.0 · November 2026

Quick start

1	Plug the USB cable into the back. The display lights white and shows **READY**.
2	Pull the handle to open the door. Put the item inside. Push the door shut until it clicks.
3	Press **TIME** to set the cook time. Press **POWER** to choose HI, MED or LO.
4	Press **START**. The display turns amber and counts down.
5	At the end the display shows **DONE** and turns green. Open the door.

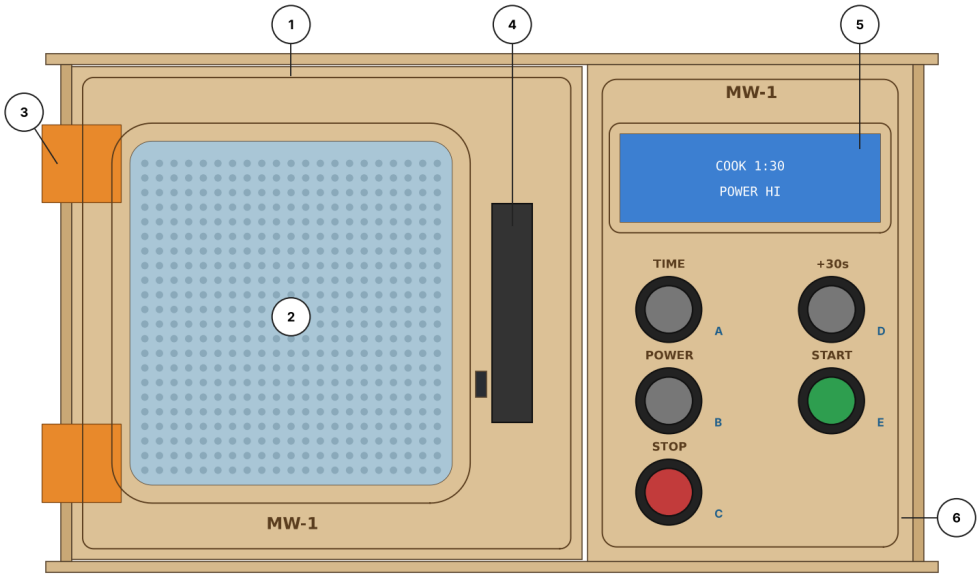
Shortcut: press **+30s** at any time to add 30 seconds. If the oven is idle it starts at once on HI.

1 Safety

NOTICE	The MW-1 is a mock-up. It has no magnetron and does not heat. Never put food you intend to eat, liquids or metal inside.
CAUTION	Keep fingers clear of the hinge side when closing the door. The door closes against the inner wall and can pinch.

- Use the USB supply only (5 V DC, 0.5 A or more). There is no mains wiring.
- Do not hang anything from the open door. It is held by two printed hinges.
- Keep the vents on the right side clear. The electronics sit behind them.

2 Product description



Front view.

No.	Part	What it does
1	Door	Opens to the left on two hinges, up to about 130°.
2	Window	Clear acrylic with a dot screen, so you can see inside.
3	Hinges	Two hinges on the left side.
4	Handle	Pull here to open. The door latches with a click when shut.
5	Display	16 × 2 character display with a coloured backlight.
6	Control panel	Five push buttons, A to E.

Control panel

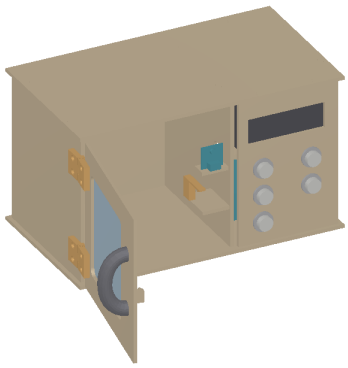
	Button	Function
A	TIME	Adds cook time: +10 s per press up to 1:00, then +1 min per press, up to 30:00.
B	POWER	Cycles power level: HI → MED → LO → HI. Shown on the second line of the display.
C	STOP	Pauses cooking. Press again to clear the time and return to READY.
D	+30s	Adds 30 s. If the oven is idle with no time set, it starts a 30 s cook on HI.
E	START	Starts or resumes cooking. Works only with the door shut and a time set.

3 Installation

1. Place the MW-1 on a flat, level surface. Leave 30 mm clear on the right side for the vents.
2. Leave the area in front clear so the door can swing open fully to the left.
3. Connect the USB cable to the socket behind the back panel opening, then to a USB port or 5 V adapter.
4. The display shows **READY** with a white backlight. If it stays blank, see Troubleshooting.

4 Operation

Opening and closing the door



Door open.

1. Grip the handle and pull straight toward you. Expect a light resistance, then a click as the latch lets go.
2. Swing the door open to the left.
3. To close, push the door shut near the handle until it clicks. The display clears any **DOOR** message.

Note: the door is detected by a sensor inside. The oven will not start or keep running while it is open.

Cooking

1. Close the door.
2. Press **TIME** until the display shows the time you want.
3. Press **POWER** to choose HI, MED or LO. The default is HI.
4. Press **START**. The backlight turns amber and the time counts down.
5. To pause, press **STOP** or open the door. Press **START** to resume.
6. When the time reaches 0:00 the display shows **DONE** and turns green for 10 s, then returns to READY.

Sounds

Sound	When
One short beep	Any button is pressed.
Three beeps	Cooking has finished (DONE).
Two low beeps	START was refused: the door is open or no time is set.

5 Display messages

Display	Backlight	Meaning
READY	White	Idle, door shut, no time set.
COOK m:ss / POWER xx	White	Time and power set, waiting for START.
m:ss counting down	Amber	Cooking.
PAUSED	Amber, steady	STOP pressed during cooking. START resumes, STOP clears.

Display	Backlight	Meaning
DOOR OPEN	Red	Door opened during cooking. Close it and press START to resume.
DONE	Green	Cook finished.
SET TIME	White	START pressed with no time set.
CLOSE DOOR	Red	START pressed with the door open.

6 Troubleshooting

Problem	Check
Display blank	USB cable seated? Try another port. Check the display's 4-wire cable inside the electronics bay.
Display lit but no text	Display contrast or wiring. The display needs 5 V.
START does nothing	Is the door fully shut (you heard the click)? Is a time set?
Door won't stay shut	The latch arm may be bent or the catch loose. See Care.
No beep when a button is pressed	The sound may be muted by a loose buzzer cable in the electronics bay. The oven still works.
One button does nothing	Check its two wires behind the control panel.

7 Care and maintenance

- Wipe with a dry or slightly damp cloth. No solvents on the acrylic window.
- Do not oil the hinges.
- If the door no longer clicks shut, the printed latch on the shelf inside the oven can be replaced. It is glued in place.

8 Technical data

Item	Value
Outside size (W × D × H)	244 × 138 × 142 mm; 163 mm deep including the handle
Inside (cooking space)	135 × 121 × 136 mm
Door	opens left, about 130°; snap latch, about 4 N to pull open
Power	5 V DC via USB, under 0.5 A
Display	Grove LCD RGB Backlight, 16 × 2 characters
Controls	5 momentary push buttons, 16 mm
Sound	Buzzer: key beep, end-of-cook signal
Materials	3 mm plywood, 3 mm clear acrylic, PLA
Heating	None

9 Warranty and service

This is a mock appliance built for MakeCU 2026. No warranty. For questions, contact the team.